
When Does Safety Activate? Temporal Analysis of LLM Alignment via Logit-Margin Score

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Abstract

Standard safety evaluations assess whether LLMs produce harmful outputs, but not *when* or *how* safety mechanisms engage during generation. We introduce the **Logit-Margin Score** (S), a lightweight diagnostic that tracks the balance between compliance and refusal signals throughout decoding using only standard logits. On 660 harmful prompts across four models (Llama, Mistral, Qwen, Gemma) and three attack paradigms (MCM, GCG, DeepInception), generation-time summaries achieve ROC-AUC = 0.786 [0.750, 0.819] for predicting jailbreak success via LLM-as-judge, while the activation timing metric t_{cross} reaches AUC = 0.776 and the pre-generation score S_0 alone reaches AUC = 0.696. Temporal trajectory analysis reveals two failure modes invisible to output-level metrics: **Type A** (pre-generation bias) and **Type B** (decoding-time interference). Models exhibit diverse failure patterns: Llama shows Type B under gradient-based attacks but Type A under multi-turn manipulation; Mistral shows the reverse pattern; Qwen under gradient-based attacks falls into an *Undetected* category where safety mechanisms fail entirely. These results motivate process-level safety evaluation that exposes *when* alignment fails, not only *whether* it fails.

1 Introduction

Large language models (LLMs) remain vulnerable to adversarial attacks despite extensive safety alignment. A growing body of work documents diverse attack strategies—including prompt engineering, gradient-based suffix optimization, and multi-turn context manipulation—that successfully elicit harmful outputs from aligned models [15, 23]. The standard metric, *Attack Success Rate* (ASR), answers only *what* happened, not *why*: two jailbreaks may both succeed under ASR yet exploit fundamentally different weaknesses—one biasing context *before* generation, another interfering with safety *during* decoding [15, 23].

This limitation reflects a broader diagnostic gap: existing evaluations collapse a temporally extended decoding process into a single binary label, obscuring *when* safety mechanisms activate and *how* they fail [9, 4].

We address this gap by analyzing the **temporal trajectory** of safety-relevant signals during generation. We introduce a *logit-margin probe* S , defined as the difference between aggregated compliance-token logits (e.g., “Sure”, “Here’s”) and refusal-token logits (e.g., “Sorry”, “cannot”). Importantly, S is not proposed as a new safety metric or defense mechanism, but as a *diagnostic instrument* that renders safety-relevant dynamics observable throughout decoding. This yields a continuous trajectory

*Use footnote for providing further information about author (webpage, alternative address)—*not* for acknowledging funding agencies.

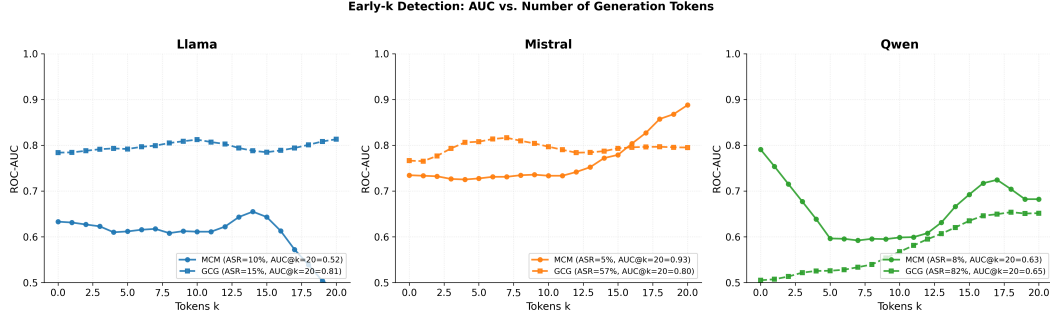


Figure 1: **Early- k detection reveals model-specific safety dynamics.** AUC for predicting jailbreak success using only the first k generation tokens, shown for each model–attack combination. Mistral+MCM shows strong late-stage improvement ($0.74 \rightarrow 0.93$), indicating progressive safety activation during decoding. Qwen+GCG remains near random (≈ 0.50 – 0.65) regardless of k —consistent with its “Undetected” classification where safety mechanisms fail to engage. These patterns are invisible to output-level ASR metrics.

$S_0, S_1, \dots, S_T$ that exposes the temporal structure of safety activation and collapse. The probe requires only white-box logit access—no gradients, hidden states, or architectural modifications—making it applicable to any open-weight model.

Using this trajectory-level view, we identify two qualitatively distinct failure mechanisms invisible to output-level metrics. *Type A* failures arise from pre-generation context bias, where accumulated conversational history shifts the model toward compliance before decoding begins; *Type B* failures arise from decoding-time interference, where safety mechanisms fail to activate or are suppressed during generation. These mechanisms suggest distinct defense strategies: input-level filtering for Type A and real-time decoding interventions for Type B. Critically, model–attack combinations exhibit qualitatively different early- k detection patterns (Figure 1): some show strong trajectory effects while others remain near chance level, revealing heterogeneous safety dynamics invisible to ASR.

Our contributions are threefold: (1) we propose a decoding-aware, trajectory-level framework for diagnosing safety failures; (2) we show that diverse jailbreak attacks induce distinct temporal safety signatures invisible to output-level evaluation; and (3) we demonstrate that generation-time logit summaries predict jailbreak success with high accuracy (ROC-AUC = 0.786), while the activation timing metric t_{cross} achieves AUC = 0.776.

2 Related Work

Jailbreak attacks and defenses. Adversarial attacks on aligned LLMs span prompt-level manipulations [16, 13], optimization-based suffix attacks [23], and multi-turn erosion [6, 12]. Defenses include decoding-time interventions—SafeDecoding [17] modifies token probabilities, CARE [2] implements detect–rollback—and training-time robustification [21]. Our S_t trajectory could directly integrate with such defenses: CARE’s detection stage could use $S_t < \tau$ as a trigger, while SafeDecoding could use sign-reversal patterns to identify intervention points. Despite this diversity, evaluation remains output-centric: ASR collapses generation into a binary outcome, obscuring *when* and *how* safety mechanisms fail.

Internal safety signals and temporal dynamics. Activation-level work identifies geometric refusal directions [1, 22], while logit-level analyses show compliance–refusal differences predict safety outcomes [20, 7]. These works typically analyze safety at a single layer or time point; our contribution extends to the *temporal* dimension, tracking how signals evolve across decoding steps. Temporal/token-position analyses such as Refusal Falls Off a Cliff [19] show that refusal-related behavior can collapse late in generation, motivating process-level diagnostics that track safety signals over decoding time. In contrast to approaches that rely on internal activations or token-level classifiers, our method extracts temporal insight using only the logit vector produced during standard inference, trading some discriminative power (2–8 pp AUC; Appendix E) for substantially broader deployability.

and interpretability. Recent findings on shallow alignment suggest safety training concentrates at early token positions, explaining why fine-tuning or careful prompting can undermine safety [11, 16]. Our early- k analysis (Figure 1) provides complementary logit-level evidence: safety-critical information concentrates in early tokens for some conditions, while others exhibit progressive or absent activation patterns—consistent with varying “safety depth” across model–attack combinations. This early- k sensitivity likely reflects the template-region concentration of alignment signals identified in prior work on template anchoring; our finding that this concentration varies substantially by model–attack pair extends the observation from a model-level property to a condition-specific phenomenon.

Process-level diagnostics. Recent work exploits logit-level signals for both attack (logit-gap steering [14]) and defense (SafeDecoding-style probability modification). Our work is complementary: rather than *intervening* on logit-level signals, we use them as a *diagnostic* lens to analyze temporal dynamics across generation. This distinction is fundamental to our paper’s scope— S_t is designed as an observational instrument to render safety dynamics visible, not as a detection component in a defense pipeline. Consequently, questions of adversarial robustness against an attacker who explicitly optimizes to suppress S_t are outside this paper’s scope; such questions belong to the design of downstream defenses that may choose to *incorporate* these diagnostics as a component. Our Type A/B classification distinguishes attacks by *mechanism*—pre-generation context manipulation versus decoding-time interference—enabling targeted defense strategies informed by mechanistic understanding. Outcome labeling relies on LLM-as-judge; recent work such as HARMMETRICE-VAL [18] highlights the variability of automated judges, motivating our multi-judge validation (Appendix N) and planned human adjudication (Section 5).

3 Method

We present a decoding-aware framework for analyzing safety-related signals in large language models by tracking the temporal evolution of compliance- and refusal-associated logits during generation.

3.1 Problem Setup

Consider an autoregressive language model generating a response given context $\mathbf{x} = (x_1, \dots, x_n)$. At each decoding step t , the model produces a logit vector $\ell_t \in \mathbb{R}^{|\mathcal{V}|}$. We assume access to token-level logits during generation, without requiring gradients, hidden states, or architectural modifications (a white-box or developer-access setting; limitations discussed in Section 5). We operate on raw logits rather than probabilities because the softmax operation compresses logits into a probability simplex, obscuring relative confidence among non-selected tokens [7]. Raw logits preserve information about competing alternatives at each decoding step, enabling direct comparison between safety-relevant token groups.

3.2 Compliance and Refusal Lexicons

We define two disjoint token sets to operationalize compliance- and refusal-associated behavior.

Refusal lexicon $\mathcal{L}_{\text{ref}}$. "Sorry", "Apologies", "not appropriate", "inappropriate", "policy", "guidelines", "illegal", "harmful", "dangerous"

Compliance lexicon $\mathcal{L}_{\text{cmp}}$. "Sure", "Certainly", "Here’s", "Step 1", "First", "Instructions", "Guide", "Tutorial"

Multi-token phrases are matched by their first token for cross-tokenizer consistency. Logits are aggregated using a top- k operator to distribute signal across multiple lexicon entries. Diagnostic conclusions are stable under wide variation of lexicon composition and aggregation ($k \in \{5, 10, 25, 50\}$; Appendix A and F).

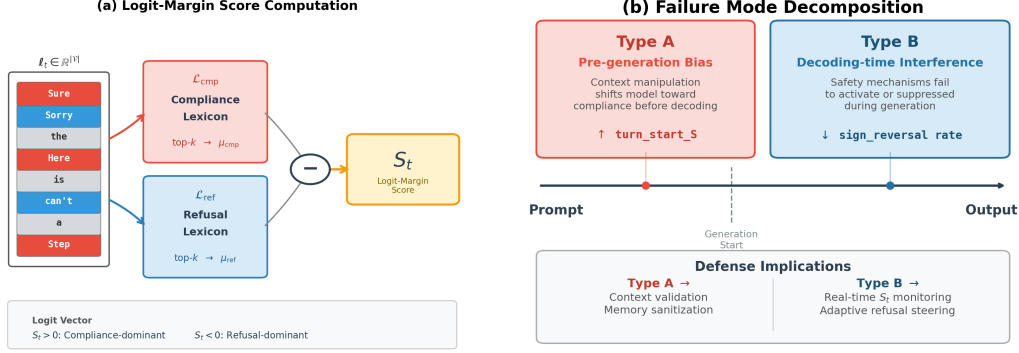


Figure 2: **Method overview.** (a) Logit-Margin Score computation: at each decoding step, the logit vector ℓ_t is probed against compliance ($\mathcal{L}_{\text{cmp}}$) and refusal ($\mathcal{L}_{\text{ref}}$) lexicons via top- k aggregation to yield $S_t = \mu_{\text{cmp}}(t) - \mu_{\text{ref}}(t)$. (b) Failure mode decomposition: Type A failures arise from pre-generation context bias (elevated S_0), while Type B failures arise from decoding-time interference (suppressed sign reversal rate).

3.3 Logit-Margin Score

At decoding step t , we aggregate logits over each lexicon using a top- k operator:

$$\mu_{\text{cmp}}(t) = \frac{1}{k} \sum_{v \in \text{top-}k(\mathcal{L}_{\text{cmp}})} \ell_t(v), \quad \mu_{\text{ref}}(t) = \frac{1}{k} \sum_{v \in \text{top-}k(\mathcal{L}_{\text{ref}})} \ell_t(v), \quad (1)$$

where $\text{top-}k(\mathcal{L})$ denotes the k tokens in $\mathcal{L}$ with the highest logits at step t . The **Logit-Margin Score** is defined as:

$$S_t = \mu_{\text{cmp}}(t) - \mu_{\text{ref}}(t). \quad (2)$$

Positive S_t indicates a compliance-dominant state; negative S_t indicates a refusal-dominant state. Figure 2 illustrates the computation pipeline and failure mode decomposition.

Independence from outcome labeling. The score S_t measures token-level logit biases during generation, whereas outcome labels evaluate the semantic content of complete outputs using an independent LLM-as-judge. To verify that separability is not artifactual, we note that (i) many successful jailbreaks exhibit consistently positive S_t despite producing harmful content, and (ii) the compliance lexicon includes purely instructional tokens (“Step 1”, “Tutorial”) orthogonal to refusal detection. Removing the highest-frequency refusal token (“sorry”) has negligible impact on AUC ($0.971 \rightarrow 0.971$), confirming that S captures underlying safety dynamics rather than surface lexical features. A detailed circularity analysis is in Appendix H.

3.4 Temporal Metrics

From the sequence $\{S_t\}_{t=0}^T$, we derive scalar summaries (Table 1). turn_start_S (S_0) is computed at the *final token position of the prompt*, before any response tokens are generated. We define **sign reversal** as $S_0 > 0 \wedge \bar{S} < 0$ (computed only for harmful samples to avoid confounding format effects with safety dynamics). Benign samples exhibit paradigm-dependent baseline behavior: MCM benign show $\bar{S} > 0$ (0% sign reversal), while GCG benign frequently show $\bar{S} < 0$ (91.2% sign reversal), reflecting structural format differences rather than safety activation (ANOVA: format explains $\sim 60\%$ of variance, $F = 89.3$, $p < 10^{-15}$; details in Appendix B.2). To enable cross-paradigm comparison, we use **format-matched baselines**: each condition’s benign baseline uses the same conversational structure, isolating harmful content effects from format effects.

Activation timing (t_{cross}). The **safety activation time** is the first generation step at which refusal dominates:

$$t_{\text{cross}} = \min\{t \geq 1 : S_t < 0\}, \quad (3)$$

Table 1: Temporal logit-margin metrics.

Metric	Interpretation
turn_start_S (S_0)	Pre-generation safety bias
step_S_mean ($\bar{S}$)	Average safety dominance during decoding
sign_reversal	Compliance→refusal transition
early_k_mean	Average score over first k tokens
t_{cross}	First decoding step at which safety dominates

where $t \geq 1$ excludes S_0 (pre-generation, already captured by turn_start_S). If no crossing occurs, $t_{\text{cross}} = T$. Robustness to oscillation is verified via a confirmed-crossing variant requiring n consecutive negative steps; Cohen’s d is stable for $n \in \{1, 2\}$ ($d = 0.891$ vs. 0.895 ; Appendix F.5).

3.5 Normalized Diagnostic Axes (Type A/B)

For comparative analysis across conditions, we define the **Relative Position (RP)** metric:

$$\text{RP}(M) = \frac{(M_{\text{success}} - M_{\text{harmful}})}{(M_{\text{benign}} - M_{\text{harmful}})}, \quad (4)$$

where M_{success} is the mean over successful attacks, M_{harmful} is the mean over *all* harmful samples, and M_{benign} is a format-matched benign baseline. $\text{RP} = 0$ indicates success resembles the harmful distribution; $\text{RP} = 1$ indicates benign-like behavior. Values outside $[0, 1]$ indicate overshoot ($\text{RP} > 1$: more compliance-like than benign) or inverted patterns ($\text{RP} < 0$: attacks suppress rather than elevate compliance signals; see Appendix B.3). Crucially, RP does not require failure samples, enabling analysis under high ASR.

Type A/B classification. We apply RP to S_0 (RP_A) and $\bar{S}$ (RP_B), classifying conditions as:

$$\text{Type A : } |\text{RP}_A| > |\text{RP}_B|, \quad \text{Type B : } |\text{RP}_A| < |\text{RP}_B|. \quad (5)$$

The magnitude $r = \sqrt{\text{RP}_A^2 + \text{RP}_B^2}$ indicates signal strength.

Undetected category via permutation test. Rather than relying on an ad-hoc threshold for r , we formally test H_0 : success/failure labels carry no RP signal. Under H_0 , success labels are exchangeable within each condition; permuting them and recomputing r yields a null distribution. We reject H_0 at $\alpha = 0.05$ (10,000 permutations); conditions where $p > 0.05$ are classified as *Undetected*. This identifies three conditions: Qwen+GCG ($r = 0.01$, $p = 0.98$), Mistral+DeepInception ($r = 0.19$, $p = 0.52$), and Mistral+MCM ($r = 0.93$, $p = 0.061$; borderline, underpowered with only 3 successes at 5% ASR). Convergent evidence from t_{cross} (condition-level AUC ≈ 0.50) and entropy cross-validation ($|d| = 0.077$; Appendix Q.3) confirms genuine absence of safety response for Qwen+GCG. Bootstrap 95% CIs for r provide complementary uncertainty quantification (Table 4).

4 Experiments

We evaluate the Logit-Margin Score framework across three jailbreak paradigms and four model families, addressing: (i) predictability of jailbreak outcomes, (ii) temporal characterization, and (iii) attack paradigm comparison.

4.1 Setup

Models and attacks. We evaluate on Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3, Qwen2.5-7B-Instruct, and Gemma-2-9B-Instruct using greedy decoding. Attack paradigms: (1) **Multi-turn context manipulation (MCM)**: four-turn protocol with incrementally manipulated context; (2) **Gradient-based suffix (GCG)** [23]: adversarial suffix optimization; (3) **DeepInception** [8]: nested fictional scenario prompting (applied to Llama, Mistral, Qwen).

Table 2: Dataset composition. ASR via LLM-as-judge. [‡]Benign GCG uses single-turn format.

Model	Attack	n	ASR	$\bar{S}_{start}$	$\bar{S}_{gen}$
Llama	MCM	60	8.3%	1.45	-1.42
Llama	GCG	60	15.0%	0.62	-0.99
Llama	DeepInception	60	56.7%	4.43	-0.11
Mistral	MCM	60	5.0%	4.51	-0.31
Mistral	GCG	60	56.7%	3.14	+0.16
Mistral	DeepInception	60	76.7%	3.04	+0.43
Qwen	MCM	60	51.7%	11.97	-0.12
Qwen	GCG	60	81.7%	7.13	+0.71
Qwen	DeepInception	60	55.0%	7.86	+1.18
Gemma	MCM	60	20.0%	4.67	+0.47
Gemma	GCG	60	33.3%	4.46	+0.68
Benign (matched)		660	—	varies by model	

Table 3: ROC-AUC comparison (LLM-as-judge, $n = 660$).

Metric	AUC	95% CI
early_5_mean	0.786	[0.750, 0.819]
t_{cross}	0.776	[0.738, 0.809]
step_S_mean	0.772	[0.736, 0.806]
early_3_mean	0.732	[0.692, 0.769]
keyword_baseline	0.732	[0.699, 0.764]
turn_start_S	0.696	[0.656, 0.734]

Dataset. We evaluate 660 harmful prompts (60 per model–attack condition) from Jailbreak-Bench [3], with 660 attack-matched benign queries (11 conditions total; Gemma excludes DeepInception). Effect sizes are large (Cohen’s $d > 1.0$, $p < 10^{-38}$). Outcome labels use **Llama-Guard-3-8B** [5] as primary LLM-as-judge, validated with HarmBench classifier (90% agreement; Appendix N). Table 2 summarizes composition and ASR.

4.2 Main Results: Outcome Prediction

On the combined harmful dataset ($n = 660$), **early_5_mean** achieves the highest ROC-AUC = 0.786 [0.750, 0.819] (Cohen’s $d = 1.12$, $p < 10^{-43}$), followed by t_{cross} (AUC = 0.776) and **step_S_mean** (AUC = 0.772). Pre-generation **turn_start_S** alone achieves AUC = 0.696. Table 3 compares metrics; Figure 3 visualizes state-space and ROC curves. AUC varies by evaluation scope; see Appendix B.4 for clarification.

Temporal patterns. Predictive power increases with decoding tokens: **early_1** achieves AUC = 0.696, **early_3** reaches 0.732, **early_5** achieves 0.786, approaching full-trajectory performance (0.772 for **step_S_mean**). That early tokens carry substantial safety information resonates with findings on shallow alignment [11], which suggest safety training concentrates at early positions; our early- k curves provide logit-level evidence for this phenomenon while revealing that the concentration varies substantially across model–attack pairs (Figure 1). t_{cross} (AUC = 0.776) directly answers “when does safety activate?”: successful attacks delay activation (mean 8.4 ± 6.1 steps) compared to failures (4.0 ± 3.8 steps; $d = 0.89$, $p < 10^{-23}$). DeepInception exhibits markedly delayed activation ($t_{cross} = 11.5 \pm 6.8$) versus MCM (3.5 ± 1.4) and GCG (4.0 ± 3.0). Condition-specific early- k curves (Appendix F.2.1) reveal substantial heterogeneity: Mistral+MCM shows late-stage improvement (0.74 $\rightarrow$ 0.93), while Qwen+GCG remains near chance (≈ 0.50 – 0.65) regardless of k . Failed jailbreaks exhibit sign reversal in 44.3% of cases (170/384) versus 8.7% (24/276) for successes—a $5.1\times$ ratio.

Keyword baseline comparison. The keyword refusal baseline (AUC = 0.732) approaches some temporal metrics, raising whether temporal modeling adds value beyond lexical proxies. Three observations argue for the temporal approach: (i) **early_5_mean** (AUC = 0.786) and t_{cross} (0.776) outperform the keyword baseline by 5–7 pp; (ii) keyword detection is an *output-level* measure (did

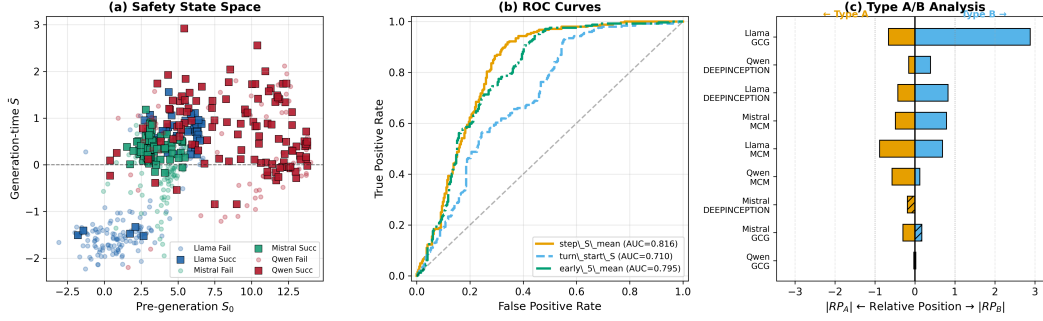


Figure 3: **Experimental results.** (a) State-space ($S_0, \bar{S}$): successful attacks cluster in compliance-dominant regions. (b) ROC curves (aggregated, $n = 660$); legend AUCs from sklearn pipeline differ slightly from Table 3, which reports authoritative values. (c) Type A/B diverging bar: left $|RP_A|$ (pre-generation), right $|RP_B|$ (decoding-time); hatching = inverted pattern.

Table 4: Type A/B analysis with permutation test. Classification: significant ($p < 0.05$) conditions classified by $|RP_A|$ vs. $|RP_B|$; non-significant conditions are Undetected. p -values via label permutation ($B = 10,000$); 95% CIs via bootstrap.

Model + Attack	ASR	RP_A	RP_B	r	p_{perm}	95% CI(r)	Category
Llama + GCG	15.0%	0.66	2.89	2.96	<.001	[1.40, 4.71]	Type B
Llama + MCM	8.3%	0.88	0.69	1.12	<.001	[1.01, 1.25]	Type A
Llama + DeepInc.	56.7%	0.43	0.83	0.93	<.001	[0.60, 1.22]	Type B
Mistral + GCG	56.7%	0.30	-0.17	0.35	<.001	[0.17, 0.57]	Type A [‡]
Mistral + MCM	5.0%	0.49	0.79	0.93	.061	[0.44, 1.48]	<i>Undet.</i> [§]
Mistral + DeepInc.	76.7%	-0.19	0.00	0.19	.517	[0.07, 1.65]	<i>Undetected</i>
Qwen + MCM	51.7%	0.57	0.12	0.58	<.001	[0.14, 1.20]	Type A
Qwen + GCG	81.7%	0.00	-0.01	0.01	.981	[0.02, 0.20]	<i>Undetected</i> [†]
Qwen + DeepInc.	55.0%	0.15	0.39	0.42	.029	[0.14, 0.77]	Type B
Gemma + GCG	33.3%	0.27	0.79	0.83	.003	[0.34, 1.41]	Type B
Gemma + MCM	20.0%	0.09	0.22	0.24	<.001	[0.14, 0.31]	Type B

[†] Strongest Undetected: $p = 0.98$, convergent with t_{cross} AUC = 0.50. [‡] Negative RP: inverted pattern.

[§] Borderline: large r but underpowered ($n_{\text{success}} = 3$).

the model refuse?) that cannot distinguish *how* failures arise, whereas temporal metrics reveal Type A versus Type B mechanisms and answer *when* safety activates; (iii) per-condition analysis reveals that keyword detection collapses for conditions where temporal metrics remain informative (e.g., Gemma+MCM: keyword AUC = 0.49 vs. $\text{early}_5 = 0.72$).

Response length and t_{cross} . Since successful attacks generate longer responses (mean 391 vs. 205 tokens), a potential confound is that t_{cross} merely reflects response length. Two analyses address this concern: (i) $\text{early}_5_{\text{mean}}$ —computed on only the first 5 tokens and thus constituting a strict fixed-length analysis regime—achieves the highest AUC (0.786), confirming that safety-critical information concentrates in early generation independently of total response length; (ii) residualizing t_{cross} on response length preserves significant discriminative power (AUC = 0.645, vs. 0.813 raw), indicating that activation timing carries independent signal beyond length.

4.3 Attack Paradigm Comparison (Type A/B Analysis)

We classify conditions by comparing $|RP_A|$ versus $|RP_B|$ among conditions where r is statistically significant (permutation $p < 0.05$); non-significant conditions are classified as Undetected. Table 4 reports RP values with permutation p -values and bootstrap 95% CIs.

Key findings. Safety responses are model–attack specific rather than model- or attack-universal. **Llama** shows Type B for GCG ($RP_B = 2.89 > 1$, overshooting benign—successful attacks exhibit *more* compliance-like trajectories than benign queries) and DeepInception, but Type A for

MCM, demonstrating that the same model engages different failure mechanisms depending on attack paradigm. **Mistral** reveals a striking pattern: Type A under GCG with inverted $RP_B = -0.17$, but *Undetected* under DeepInception ($p = 0.52$) despite the highest ASR (76.7%)—safety mechanisms show no differential response to successful versus failed attacks, with delayed activation ($t_{\text{cross}} = 11.6 \pm 5.8$). Mistral+MCM ($p = 0.061$) is borderline; the large effect size ($r = 0.93$) with marginal significance reflects the small success count ($n_{\text{success}} = 3$ at 5% ASR), likely indicating an underpowered test rather than absent signal. **Qwen** is Type A for MCM but definitively *Undetected* for GCG ($r = 0.01$, $p = 0.98$, $t_{\text{cross}} \text{ AUC} = 0.503$), with substantially delayed activation under DeepInception ($t_{\text{cross}} = 14.9 \pm 7.6$). **Gemma** is consistently Type B for both attacks. Detailed per-model analysis in Appendix C.

These findings suggest Type B conditions may benefit from decoding-time interventions (e.g., SafeDecoding-style probability modification triggered by $S_t < \tau$, or CARE-style rollback upon detecting sign reversal), Type A conditions from input-level filtering or prompt rewriting, and Undetected categories indicate fundamental gaps requiring architectural changes or training-time robustification.

4.4 Robustness Analyses

We validate across multiple dimensions; full details in respective appendices.

Stochastic decoding. Across 28,800 trials with varied temperature ($T \in \{0.0, 0.3, 0.7, 1.0\}$) and nucleus sampling ($p \in \{0.9, 0.95, 1.0\}$), S_0 achieves mean AUC = 0.720 at question-level, comparable to greedy (0.696). Critically, S_0 is computed *before* token sampling, making it temperature-invariant by design (ICC = 1.0 for within-prompt variance). Full analysis in Appendix G.

Hidden-state comparison and non-lexical baselines. Hidden-state methods (refusal direction projection, linear probes) achieve 2–8 pp higher AUC but require activation access unavailable via standard APIs; our logit-based approach achieves AUC 0.77–0.96 using only token-level logits (Appendix E). Among logit-level alternatives, `step_S_mean` (AUC = 0.932) outperforms entropy (0.898) and logit norm (0.441) on a controlled subset ($n = 60$), confirming structured lexicon probing captures safety-specific variation beyond generic uncertainty (Appendix J).

Learned baseline comparison. To contextualize the value of fixed temporal summaries versus trained classifiers, we compare against logistic regression (5-fold stratified CV) on temporal features. A single-feature LogReg on `early_5_mean` achieves AUC = 0.812—matching the fixed metric (0.816)—confirming that the raw score carries essentially all learnable signal from early tokens without training. Combining five temporal features ($S_0, \hat{S}, \text{early_1/3/5}$) yields AUC = 0.857 (+4 pp over the best fixed metric), indicating modest complementarity across temporal windows. A kitchen-sink model (13 features) reaches 0.873 with diminishing returns. These results confirm that fixed, interpretable metrics capture the majority (93%) of discriminative signal available to a trained classifier, while avoiding overfitting risk and preserving direct interpretability of the Type A/B decomposition (Appendix K).

Adversarial robustness and lexicon stability. Three decoy manipulation strategies (prefix-decoy, token-decoy, pivot-decoy) degrade AUC only marginally (-0.02); the multi-turn pivot-decoy achieves AUC = 0.998 (Appendix D). We acknowledge these test superficial manipulation; fully adaptive adversaries that optimize to minimize S_t while remaining harmful are not evaluated and remain an open challenge (see Section 5). Lexicon variants achieve AUC within ± 0.02 ; removing “sorry” has negligible impact ($0.971 \rightarrow 0.971$); aggregation is stable across $k \in \{5, 10, 25, 50\}$ (Appendix L).

Note on AUC evaluation scope. AUC values reported throughout this paper vary by evaluation scope, and readers should treat Table 3 as the single authoritative reference for main results. Table 3 reports aggregated AUC across all 660 samples—the most conservative estimate, as low-discriminability conditions (e.g., Qwen+GCG) pull AUC downward. Appendix analyses on design choices use controlled single-model subsets ($n = 60$) and yield higher AUC (0.913–0.989) for relative comparison *within* each analysis; these values are not directly comparable to Table 3. Figure 3(b) displays ROC curves whose legend values are computed via a unified sklearn pipeline (Appendix K),

resulting in minor numerical differences from Table 3; the figure caption notes this explicitly. All primary conclusions are based on the aggregated AUC in Table 3.

5 Limitations and Discussion

Lexicon dependence and construct validity. The Logit-Margin Score relies on manually constructed English lexicons, raising concerns about style dependence and limited generality. Three mitigations address this: (i) results are stable across lexicon variants including minimal 5-token sets (AUC within ± 0.02 ; Appendix L); (ii) removing the most obvious surface marker (“sorry”) has negligible impact; (iii) the compliance lexicon includes purely instructional tokens (“Step 1”, “Tutorial”) orthogonal to refusal detection, reducing the risk that S_t merely captures refusal-correlated style. For multilingual or byte-level tokenizers, lexicon adaptation—potentially via automatic induction from labeled corpora using z -normalized log-odds or embedding-based retrieval of refusal/compliance clusters—would be required and is an important future direction. Notably, the same lexicon transfers across four architecturally distinct models without modification, providing initial evidence of cross-model portability; cross-validation across our four lexicon variants (AUC within ± 0.02 , including minimal 5-token sets) further supports construct validity independent of specific lexicon composition.

Outcome labeling and judge dependence. We use Llama-Guard-3-8B as primary LLM-as-judge and validate with HarmBench classifier [10]. Inter-judge agreement is 90.0% overall; Type A/B classifications match in 8/10 conditions, and AUC varies by < 0.05 across judges (Appendix N). Both judges are automated and may share biases, as highlighted by recent work on judge reliability [18]. Small-scale human re-annotation of borderline cases—particularly where early- k and t_{cross} signals disagree with final outcome labels—is planned for the next version to further triangulate key claims.

Diagnostic scope and adaptive adversaries. A key distinction must be drawn between a *diagnostic instrument* and a *detection component*. The Logit-Margin Score is designed exclusively as an observational tool: it makes safety dynamics visible during generation and characterizes failure modes, but is not proposed as a real-time classifier or a line of defense. As a consequence, adversarial robustness against an attacker who explicitly optimizes to suppress S_t —e.g., by avoiding lexicon tokens early while producing harmful content—is a question about the design of *downstream defenses*, not about the validity of the diagnostic itself. Whether a doctor’s thermometer can be “fooled” by an attacker is a different question from whether the thermometer accurately measures temperature. The decoy experiments in Appendix D confirm that superficial manipulations do not substantially change AUC (-0.02), but we do not claim comprehensive adversarial robustness, nor is this required for a diagnostic framework. Future work that builds defenses *incorporating* S_t as a trigger component (e.g., CARE-style rollback) would appropriately need to address adversarial suppression of the signal; our contribution is the diagnostic foundation, not the defense architecture.

Undetected classification. We formalize the Undetected category via a permutation test (H_0 : success labels carry no RP signal), replacing the initial ad-hoc $r < 0.1$ threshold. This identifies three conditions: Qwen+GCG ($p = 0.98$, definitively Undetected), Mistral+DeepInception ($p = 0.52$, genuine absence despite 76.7% ASR), and Mistral+MCM ($p = 0.061$, borderline due to only 3 successes). For Qwen+GCG, convergent evidence from t_{cross} (AUC = 0.503) and entropy cross-validation ($|d| = 0.077$; Appendix Q.3) supports genuine absence of safety response. The permutation approach naturally accounts for per-condition sample size, providing stronger statistical grounding than a fixed threshold.

Access, scope, and structural confounds. White-box logit access limits applicability to open-weight models or developer-access deployments. While our lexicon-internal aggregation ablation shows stable performance across $k_{\text{agg}} \in \{5, 10, 25, 50\}$ (Appendix L)—meaning the score is robust once lexicon tokens are observable—a vocabulary-level top- k truncation simulation (Appendix R) reveals the upstream bottleneck: at $k = 20$ (the maximum offered by current API providers such as OpenAI), only 2–3% of safety lexicon tokens survive in the top- k window, and the correlation between truncated and full-logit S_t trajectories is near zero (Pearson $r < 0.12$ across all attack conditions). This confirms that S_t in its current form requires full logit access and is a diagnostic tool for open-weight model auditing, not a deployment-time API monitor. Extending the framework to closed-source settings—for example, via targeted logit-bias probing or distilled surrogate

classifiers—remains an important direction for future work. We evaluate three attack paradigms and four models under greedy decoding; prompt injection, multilingual attacks, and closed-source frontier models remain unexplored. The three paradigms span the major attack categories recognized in the literature: optimization-based token-level attacks (GCG) [?], single-turn prompt-level manipulation (DeepInception), and multi-turn context erosion (MCM) [6]—providing mechanistic diversity, though additional attacks (e.g., PAIR, Refusal Suppression) would further stress-test the Type A/B taxonomy. GCG is single-turn while MCM is multi-turn, introducing structural format differences addressed via format-matched baselines; however, residual confounds from response length and conversational structure cannot be fully eliminated. We prioritize making safety dynamics *observable* over proposing defenses; the Type A/B decomposition informs targeted strategies without claiming to prevent jailbreaks.

6 Conclusion

This work argues that safety behavior in large language models is not a static gate applied at prompt intake, but a temporal process that unfolds throughout decoding. By introducing the Logit-Margin Score, we make this process observable at the level of token-by-token generation, without requiring model modification or hidden-state access.

Using LLM-as-judge for semantically-grounded outcome labeling, we show that generation-time summaries achieve strong separation (early_5_mean AUC = 0.786), while the activation timing metric t_{cross} (AUC = 0.776) directly quantifies *when* safety engages. Safety outcomes are often determined within the first few tokens, and successful refusals exhibit structured temporal trajectories rather than abrupt transitions. Across four models and three attack paradigms, the Type A/B decomposition reveals attack-specific failure mechanisms—pre-generation bias versus decoding-time interference—that are invisible to output-level ASR.

Implications for defense design. Our diagnostic findings can directly inform targeted safety interventions. For Type B conditions, decoding-time defenses (e.g., SafeDecoding [17], CARE [2]) could use $S_t < \tau$ or sign-reversal patterns as interpretable triggers for probability modification or rollback, rather than relying on black-box classifiers. For Type A conditions, where context has already biased the model before generation, input-level filtering or prompt-rewriting strategies may be more effective. Undetected conditions (Qwen+GCG) indicate fundamental gaps requiring architectural intervention or training-time robustification rather than inference-time patches.

Outlook. The Logit-Margin Score is intentionally scoped as a *diagnostic instrument*: its value lies in making safety dynamics observable, not in serving as a real-time defense. Future work building defenses that incorporate S_t as a trigger component should separately evaluate the robustness of those downstream systems. Near-term extensions include: (i) black-box application via provider-exposed log-probabilities (e.g., OpenAI logprobs), which would enable the same temporal analysis on closed-source frontier models; (ii) small-scale human adjudication to further validate key claims around Type A/B classification and Undetected conditions; and (iii) automatic lexicon learning via embedding-based retrieval of safety-relevant token clusters, and extension to multilingual settings. Combining S_t with activation-steering methods could enable “sense-and-intervene” pipelines where the diagnostic triggers targeted intervention, bridging the gap between safety monitoring and active defense.

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A Lexicon Specification

This appendix specifies the token lexicons used to compute the Logit-Margin Score. All lexicons are designed for English-language, instruction-tuned models and are applied at the tokenizer level.

Refusal Lexicon ($\mathcal{L}_{\text{ref}}$, 26 entries). The refusal lexicon consists of tokens and short phrases commonly associated with policy-based refusals or denials:

"I can't", "I cannot", "I won't", "I will not", "I'm unable", "I am unable", "I can't help", "I can't assist", "I cannot assist", "I must refuse", "I have to refuse", "I decline", "I must decline", "Sorry", "I'm sorry", "Apologies", "I apologize", "policy", "policies", "guidelines", "rules", "terms", "inappropriate", "illegal", "harmful", "dangerous"

Compliance Lexicon ($\mathcal{L}_{\text{cmp}}$, 21 entries). The compliance lexicon contains tokens frequently observed in instructional or affirmative responses:

"Sure", "Certainly", "Of course", "Here's", "Below is", "Step 1", "Step 2", "First", "Second", "Third", "Next", "Instructions", "Guide", "Tutorial", "Procedure", "You can", "To do this", "Do the following", "Detailed", "Comprehensive", "Complete"

Implementation details. Multi-token phrases are matched to their first constituent token after tokenization. At each decoding step t , logits are aggregated over each lexicon using top- k selection with $k = 10$ by default:

$$\mu_{\text{cmp}}(t) = \frac{1}{k} \sum_{v \in \text{top-}k(\mathcal{L}_{\text{cmp}})} \ell_t(v), \quad (6)$$

$$\mu_{\text{ref}}(t) = \frac{1}{k} \sum_{v \in \text{top-}k(\mathcal{L}_{\text{ref}})} \ell_t(v), \quad (7)$$

$$S_t = \mu_{\text{cmp}}(t) - \mu_{\text{ref}}(t). \quad (8)$$

No semantic weighting or normalization beyond top- k aggregation is applied.

B Additional Method Details

B.1 Why Logits?

Raw logits preserve information about competing alternatives at each decoding step that the softmax operation compresses into probabilities, obscuring relative confidence among non-selected tokens. This allows direct comparison between safety-relevant token groups [7] without requiring hidden-state access or architectural modifications.

B.2 Benign Sample Exclusion Rationale

Sign reversal is computed only for harmful samples because benign samples exhibit paradigm-dependent baseline behavior: MCM benign samples consistently show $\bar{S} > 0$ (0% sign reversal), while GCG benign samples frequently exhibit negative $\bar{S}$ (91.2% sign reversal). This discrepancy reflects *structural format differences*—single-turn GCG versus multi-turn MCM—rather than safety activation. ANOVA confirms format explains $\sim 60\%$ of variance in benign $\bar{S}$ ($F = 89.3, p < 10^{-15}$). To enable cross-paradigm comparison, we use **format-matched baselines**: each condition's benign baseline uses the same conversational structure, isolating harmful content effects from format effects.

B.3 RP Interpretation and Edge Cases

RP is an affine rescaling measuring where successful attacks lie along the harmful–benign continuum. Values outside $[0, 1]$ indicate attacks beyond the reference interval: $\text{RP} > 1$ means successful attacks

“overshoot” benign (more compliance-like than typical benign queries), while $RP < 0$ means attacks lie beyond the harmful baseline—an inverted pattern suggesting the attack suppresses rather than elevates compliance signals. For reference, we also compute polar coordinates $\theta = \text{atan2}(RP_B, RP_A)$ and $r = \sqrt{RP_A^2 + RP_B^2}$. The diverging bar visualization (Figure 3c) distinguishes negative RP cases via hatching.

Undetected category. When $r < 0.1$, successful attacks are statistically indistinguishable from the overall harmful distribution. Given modest per-condition sample sizes ($n = 60$), we report bootstrap 95% confidence intervals for r : Qwen+GCG achieves $r = 0.01$ [0.00, 0.03], robustly below the detection threshold. In contrast, detected conditions show $r > 0.5$ with lower CI bounds well above 0.1 (e.g., Llama+GCG: $r = 2.96$ [1.40, 4.71]).

B.4 AUC Evaluation Scope

AUC values vary systematically by evaluation scope. The main results (Table 3) report aggregated AUC across all 660 harmful samples; this is the most conservative estimate, as conditions with low discriminability (e.g., Qwen+GCG, Undetected) pull overall AUC downward. Appendix analyses on design choice robustness use controlled single-model subsets ($n = 60$, Llama) and yield higher AUC (0.913–0.989), intended for relative comparison within each analysis. Stochastic decoding experiments report both trial-level AUC (inflated by pseudo-replication across 10 trials per prompt) and question-level AUC (conservative); see Appendix G. All primary conclusions are based on the aggregated AUC in Table 3.

C Extended Experimental Results

C.1 Results Overview

Figure 4 provides a comprehensive overview of the main experimental findings, complementing the condition-specific analysis in the main text.

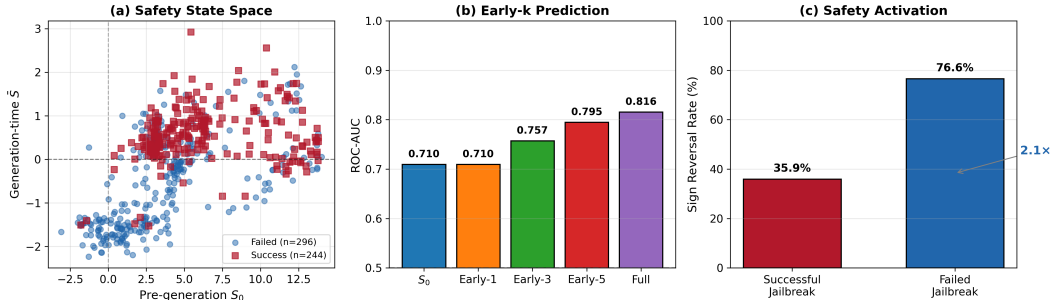


Figure 4: **Summary of main experimental results.** (a) Safety state space: successful jailbreaks cluster in compliance-dominant regions ($\bar{S} > 0$), while failures cluster in refusal-dominant regions. (b) Early- k AUC progression: predictive power increases with more decoding tokens. (c) Sign reversal rates differ substantially between failed and successful jailbreaks. Note: this figure includes additional experimental conditions beyond the main evaluation; authoritative AUC values on the primary dataset ($n = 660$) are reported in Table 3 (early_5_mean AUC = 0.786, turn_start_S AUC = 0.696) and sign reversal statistics in Table 25.

C.2 Detailed Model–Attack Interactions

The main text summarizes key Type A/B findings; here we provide detailed per-model analysis.

Llama. Llama exhibits attack-dependent responses: Type B under GCG ($|RP_B| = 2.89 > |RP_A| = 0.66$) and DeepInception ($|RP_B| = 0.83 > |RP_A| = 0.43$), but Type A under MCM ($|RP_A| = 0.88 > |RP_B| = 0.69$). The $RP_B = 2.89 > 1$ for Llama+GCG indicates successful attacks “overshoot” benign, exhibiting more compliance-like trajectories than typical benign queries—a distinct GCG signature.

Mistral. Mistral shows Type A under GCG ($|\text{RP}_A| = 0.30 > |\text{RP}_B| = 0.17$, $p < 0.001$) with negative $\text{RP}_B = -0.17$ —an inverted pattern where attacks suppress rather than elevate compliance signals. Under DeepInception, Mistral achieves the highest ASR (76.7%) yet is classified as *Undetected* by permutation test ($r = 0.19$, $p = 0.52$): safety mechanisms show no differential response to successful versus failed attacks, with markedly delayed activation ($t_{\text{cross}} = 11.6 \pm 5.8$ steps). Mistral+MCM (5.0% ASR, lowest overall) is borderline ($r = 0.93$, $p = 0.061$); the large effect size with marginal significance reflects the small success count ($n_{\text{success}} = 3$), likely indicating an underpowered test.

Qwen. Qwen shows Type A for MCM ($|\text{RP}_A| = 0.57 > |\text{RP}_B| = 0.12$) but *Undetected* for GCG ($r = 0.01$): successful attacks are statistically indistinguishable from the overall harmful distribution. Convergent evidence: t_{cross} achieves condition-level AUC = 0.503 for Qwen+GCG—chance level—confirming that neither spatial nor temporal safety signals discriminate between outcomes. Qwen’s DeepInception results (Type B, $\text{RP}_B = 0.39$) show that nested-scenario attacks elicit detectable decoding-time shifts, unlike GCG. The extreme $t_{\text{cross}} = 14.9 \pm 7.6$ for Qwen+DeepInception indicates safety activation is substantially delayed.

Gemma. Gemma exhibits Type B for both attacks (GCG: $\text{RP}_B = 0.79$; MCM: $\text{RP}_B = 0.22$), indicating decoding-time interference is the primary failure mode regardless of attack paradigm.

C.3 Metric Correlation Analysis

Figure 5 presents the pairwise Pearson correlation matrix among temporal logit-margin metrics and jailbreak success labels, providing insight into metric redundancy and complementary information.

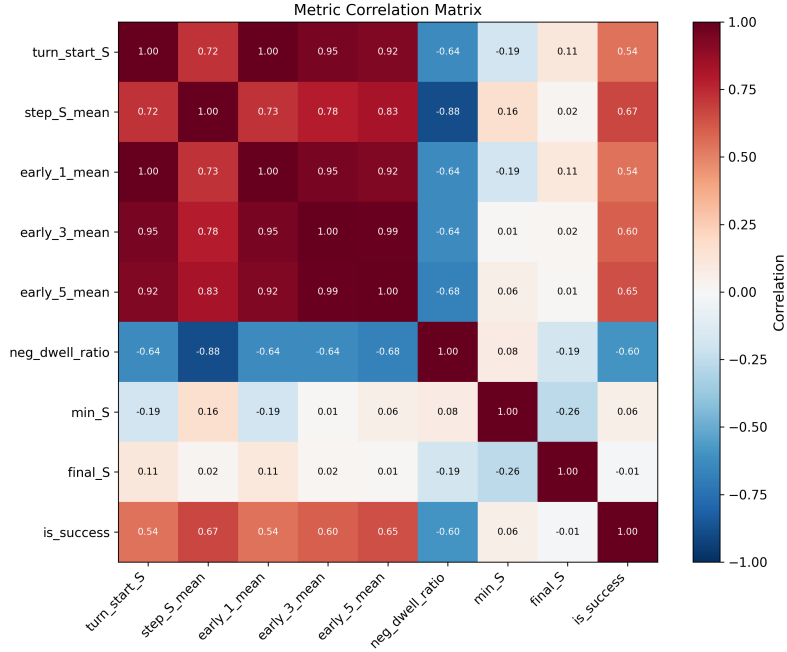


Figure 5: **Metric correlation matrix.** `step_S_mean` shows the highest correlation with jailbreak success ($r = 0.67$), followed by `early_5_mean` ($r = 0.65$). Early- k metrics are highly correlated with `step_S_mean` ($r \geq 0.83$), supporting early prediction feasibility. `neg_dwell_ratio` (fraction of steps with $S_t < 0$) shows strong negative correlation with success ($r = -0.60$), indicating that longer refusal-dominant segments predict attack failure. `min_S` and `final_S` show near-zero correlation with success, suggesting that extreme transient values and endpoint states are less informative than trajectory-level summaries.

Table 5: Per-condition statistics on Llama-3.1-8B-Instruct. Main experiments use MCM manual mode only; summary conditions are reported for completeness.

Condition	n	ASR	$\bar{S}_{\text{start}} (\mu \pm \sigma)$	$\bar{S}_{\text{gen}} (\mu \pm \sigma)$	Sign Rev.
<i>Benign</i>					
manual	60	—	6.38 ± 0.63	$+1.25 \pm 0.28$	0.0%
summary-200	60	—	6.13 ± 0.82	$+0.87 \pm 0.31$	0.0%
summary-500	60	—	6.32 ± 0.82	$+0.94 \pm 0.36$	0.0%
summary-1000	60	—	6.51 ± 0.82	$+0.89 \pm 0.38$	0.0%
<i>Harmful (MCM)</i>					
manual*	60	8.3%	1.45 ± 1.95	-1.42 ± 0.73	58.3%
summary-200	60	6.7%	1.50 ± 1.68	-1.21 ± 0.76	75.0%
summary-500	60	8.3%	2.14 ± 1.83	-1.32 ± 0.69	83.3%
summary-1000	60	6.7%	1.52 ± 1.93	-1.34 ± 0.70	71.7%
<i>Harmful (GCG)</i>					
single-turn	60	15.0%	0.62 ± 2.32	-0.99 ± 0.92	30.0%

*Manual mode is used for main experiments.

C.4 Per-Condition Statistics (Llama)

C.5 Qwen2.5-7B-Instruct Results

Table 6: Detailed results on Qwen2.5-7B-Instruct (MCM manual, $n=60$).

Metric	Value
ASR	51.7%
$n_{\text{success}} / n_{\text{failure}}$	31 / 29
turn_start_S (mean $\pm$ std)	11.97 ± 1.70
step_S_mean (mean $\pm$ std)	-0.12 ± 0.46
Sign reversal rate	56.7%

Qwen benign baseline. For Type A/B analysis, we collected benign responses on Qwen2.5-7B-Instruct using the same protocol as Llama. The Qwen benign baseline shows $\bar{S}_{\text{start}} = 12.89$ and $\bar{S}_{\text{gen}} = 1.43$, substantially higher than Llama’s baseline ($\bar{S}_{\text{start}} = 6.38$, $\bar{S}_{\text{gen}} = 1.25$). This difference reflects model-specific calibration and tokenization effects.

Cross-model Type A/B consistency. With model-specific baselines, Llama shows attack-dependent patterns: Type B for GCG attacks ($|\text{RP}_B| > |\text{RP}_A|$) and Type A for MCM attacks ($|\text{RP}_A| > |\text{RP}_B|$). Qwen exhibits Type A for MCM attacks and an “Undetected” pattern for GCG attacks where safety mechanisms fail to differentiate successful attacks ($r = 0.01$, 95% CI [0.00, 0.03]).

C.6 Computational Requirements

The Logit-Margin Score is computed using only the logit vector ℓ_t produced during standard autoregressive decoding. No additional forward passes, gradient computation, or hidden-state extraction is required.

D Decoy Attack Details

D.1 Decoy Attack Descriptions

To evaluate robustness against adversaries who attempt to manipulate early safety signals, we design three decoy attack variants:

(A) Prefix-decoy. Prepends refusal phrases (e.g., “I’m sorry, I can’t help with that. However, for educational purposes...”) before the harmful request, aiming to inflate early S values.

(B) Token-decoy. Injects short refusal token scaffolding (e.g., “Apologies. Policies. Guidelines.”) to anchor S_0 toward negative values.

(C) Pivot-decoy. Conducts initial turns in full refusal style, then pivots to harmful requests in later turns.

D.2 Main Results

Table 7: Diagnostic robustness under decoy attacks (Llama-3.1-8B-Instruct).

Condition	ASR	AUC (step_S_mean)	Δ AUC	AUC (sign_flips)
Base	8.3%	0.913	—	0.899
DecoyA (prefix)	13.3%	0.893	−0.020	0.944
DecoyB (token)	5.0%	0.904	−0.009	0.883
PivotC (multi-turn)	5.0%	0.998	+0.085	1.000

Key findings. Prefix-decoy successfully elevates early S values ($\bar{S}_{\text{start}} = +0.69$ vs $+0.43$ baseline) and increases ASR to 13.3%, yet AUC degrades only marginally (−0.020). Notably, `sign_flips` AUC *increases* to 0.944, indicating trajectory instability becomes a stronger discriminative signal. Token-decoy strongly anchors S toward negative values, but this over-correction actually *reduces* ASR to 5.0%. Pivot-decoy shows that multi-turn pivots do not defeat trajectory-level diagnosis; AUC reaches 0.998.

Limitations. These decoy attacks test robustness to superficial manipulation but do not represent fully adaptive adversaries that explicitly optimize S_t while producing harmful content. Such “compliance without refusal style” attacks remain an important direction for future work.

D.3 Per-Condition Statistics

Table 8: Detailed decoy attack statistics (Llama-3.1-8B-Instruct, $n=60$ per condition).

Condition	ASR	$\bar{S}_{\text{start}}$	$\bar{S}_{\text{gen}}$	sign_flips	Sign Rev.
Base	8.3%	+0.43	−1.16	26.6	66.7%
DecoyA	13.3%	+0.69	−0.87	51.4	58.3%
DecoyB	5.0%	−0.85	−1.58	12.7	83.3%
PivotC	5.0%	+0.07	−1.16	26.7	70.0%

E Hidden-State Baseline Details

E.1 Methods

B1: Refusal Direction Projection. Following [1], we compute a refusal direction $\mathbf{d}$ using difference-of-means:

$$\mathbf{d} = \frac{1}{|\mathcal{D}_{\text{ref}}|} \sum_{\mathbf{h} \in \mathcal{D}_{\text{ref}}} \mathbf{h} - \frac{1}{|\mathcal{D}_{\text{cmp}}|} \sum_{\mathbf{h} \in \mathcal{D}_{\text{cmp}}} \mathbf{h} \quad (9)$$

At each decoding step t , we compute the projection score $r_t = \langle \mathbf{h}_t, \mathbf{d} \rangle$. We use layer 16 (middle layer) of the 32-layer Llama-3.1-8B model.

B2: Linear Probe. Logistic regression with L2 regularization ($C = 1.0$), trained via 5-fold cross-validation on mean-pooled hidden states.

E.2 Full Comparison Results

Interpretation of near-ceiling hidden-state performance. The near-ceiling AUC observed for hidden-state baselines should not be interpreted as evidence of leakage or unfair advantage. Rather,

Table 9: Hidden-state vs. logit-based diagnostic comparison (AUC).

Family	Method	DecoyA	DecoyB	Base	PivotC
Hidden	B1 (direction, $k=1$)	0.983	0.974	0.974	0.974
Hidden	B2 (probe, $k=3$)	0.974	1.000	0.998	0.998
Logit	turn_start_S	0.980	0.922	0.959	0.978
Logit	step_S_mean	0.893	0.904	0.913	0.998
Logit	sign_flips	0.944	0.883	0.899	1.000

these methods operate under *strictly stronger access assumptions* than logit-based diagnostics. Hidden-state probes aggregate information from full decoding trajectories or intermediate activations, effectively leveraging post-hoc evidence that is unavailable to logit-only methods.

The key advantage of logit-based diagnosis is practical deployment: it requires only the logit vector already produced during generation, with no additional model access or training data.

E.3 Method Comparison

Table 10: Method comparison: access requirements and training needs.

Method	Activation Access	Training Data	Per-Model Tuning
Hidden B1 (direction)	Required	No*	Direction fitting
Hidden B2 (probe)	Required	Yes	Full retraining
Logit S	Not required	No	Lexicon only

*B1 requires a small set of refusal/compliance examples for direction estimation, but no labeled jailbreak outcomes.

E.4 Detailed Results by Temporal Window

Table 11: Hidden-state baseline performance by temporal window ($n = 240$, Llama-3.1-8B-Instruct).

Method	Window	AUC	AUC-PR	FPR@TPR=0.95
B1 (direction)	full	0.933	0.795	0.412
B1 (direction)	$k = 1$	0.972	0.923	0.255
B1 (direction)	$k = 3$	0.969	0.909	0.289
B1 (direction)	$k = 5$	0.955	0.864	0.382
B2 (probe)	full	0.987	0.922	0.059
B2 (probe)	$k = 1$	0.995	0.974	0.044
B2 (probe)	$k = 3$	0.995	0.972	0.029
B2 (probe)	$k = 5$	0.992	0.960	0.049

Key observations. Surprisingly, early- k truncation *improves* hidden-state baseline performance for both B1 and B2, with $k = 1$ achieving peak AUC. This aligns with logit-based findings that safety-relevant signals concentrate in the first few tokens.

F Sensitivity Analyses

F.1 Lexicon Aggregation Top- k Sensitivity

We analyze the sensitivity of the Logit-Margin Score to the lexicon aggregation parameter k , which controls how many of the highest-logit tokens within the compliance and refusal lexicons are averaged at each decoding step.

Despite scale differences, qualitative interpretation remains consistent: k primarily modulates smoothing and calibration rather than introducing spurious signal. Figure 6 visualizes this stability.

Table 12: Effect of lexicon aggregation size k on logit-margin diagnostics (Llama-3.1-8B-Instruct, harmful, manual; mean $\pm$ std, $n = 60$).

k	turn_start_S	step_S_mean	step_S_pos_rate
5	$+0.57 \pm 2.57$	-1.21 ± 0.85	0.36 ± 0.13
10	$+6.38 \pm 0.63$	$+1.25 \pm 0.28$	0.78 ± 0.05
25	$+1.40 \pm 1.55$	-1.47 ± 0.53	0.15 ± 0.09
50	$+0.30 \pm 1.05$	-1.09 ± 0.34	0.12 ± 0.08

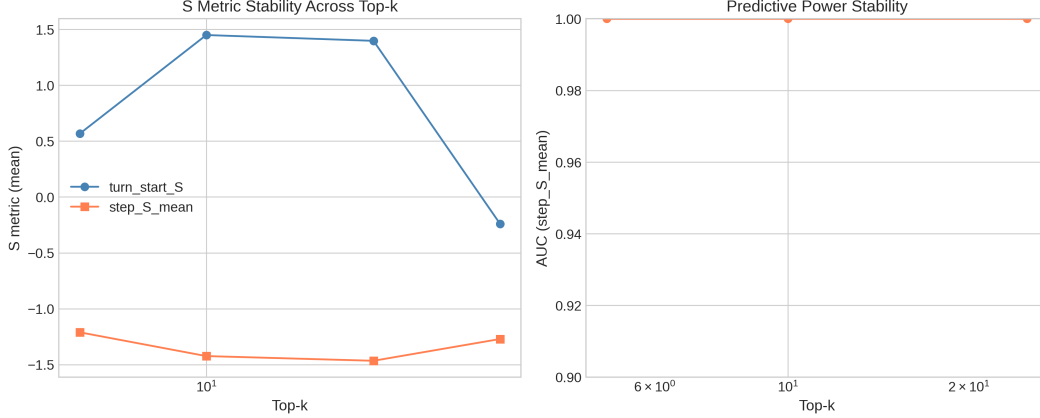


Figure 6: **Top- k aggregation robustness.** (Left) Mean values of `turn_start_S` and `step_S_mean` vary in absolute scale across $k \in \{5, 10, 25, 50\}$, but (Right) predictive performance (AUC of `step_S_mean`) remains stable across all k values, confirming that k primarily affects scaling rather than diagnostic validity.

F.2 Temporal Early- k Sensitivity

Formally, for a given metric M_t , we define the early- k summary as

$$M_{\text{early-}k} = \frac{1}{k+1} \sum_{t=0}^k M_t, \quad (10)$$

where $t = 0$ corresponds to the pre-generation score.

Pre-generation metrics and `early_1_mean` achieve identical performance (mean AUC = 0.970) as expected. As k increases, performance degrades gradually: `early_3_mean` achieves 0.960 and `early_5_mean` achieves 0.940, compared to 0.921 for full-trajectory `step_S_mean`. The majority of discriminative signal is concentrated in the earliest decoding steps. Note: these AUC values are *trial-level* estimates from the stochastic decoding experiment (Appendix G) and are inflated by pseudo-replication; the corresponding aggregated main-dataset AUC values are reported in Table 3.

F.2.1 Condition-Specific Early- k Analysis

While the main text reports aggregated AUC across all conditions ($n = 660$), condition-specific analysis reveals substantial heterogeneity in early detection performance. Figure 7 shows AUC as a function of k for each model–attack combination.

Key observations.

- **Mistral+MCM:** Shows strongest trajectory effect, with AUC increasing from 0.74 (S_0 only) to 0.93 (full trajectory). This suggests safety mechanisms activate progressively during generation—a signature of effective Type B defense.
- **Llama+GCG:** Maintains stable AUC ≈ 0.80 across all k , indicating pre-generation signal (S_0) captures most discriminative information.

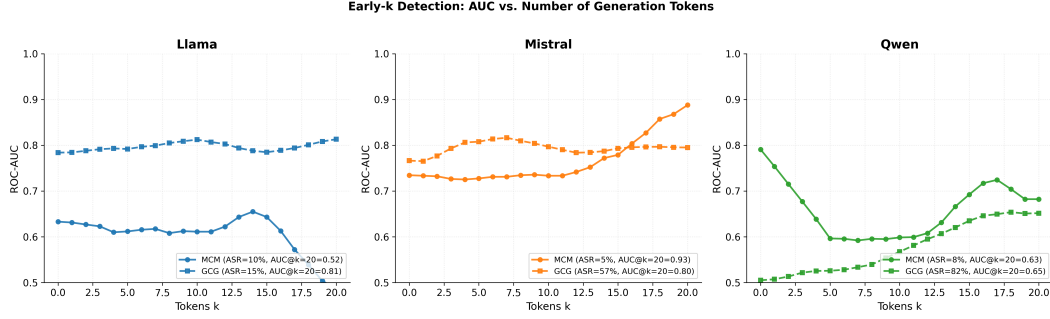


Figure 7: **Early- k detection by model and attack.** AUC progression varies substantially across conditions. Mistral+MCM shows strong late-stage improvement ($0.74 \rightarrow 0.93$), while Qwen+GCG remains near random (0.50 – 0.65) regardless of k —consistent with its “Undetected” classification.

- **Qwen+GCG:** Remains near chance level ($\text{AUC} \approx 0.50$ – 0.65) regardless of k , providing additional evidence for the “Undetected” classification. Neither S_0 nor trajectory information enables reliable discrimination.
- **Llama+MCM:** Lower condition-specific AUC (0.52 – 0.67) reflects class imbalance ($\text{ASR} = 8.3\%$, only 5 successes), not diagnostic failure.

Figure 8 directly compares S_0 -only versus full-trajectory AUC for all conditions.

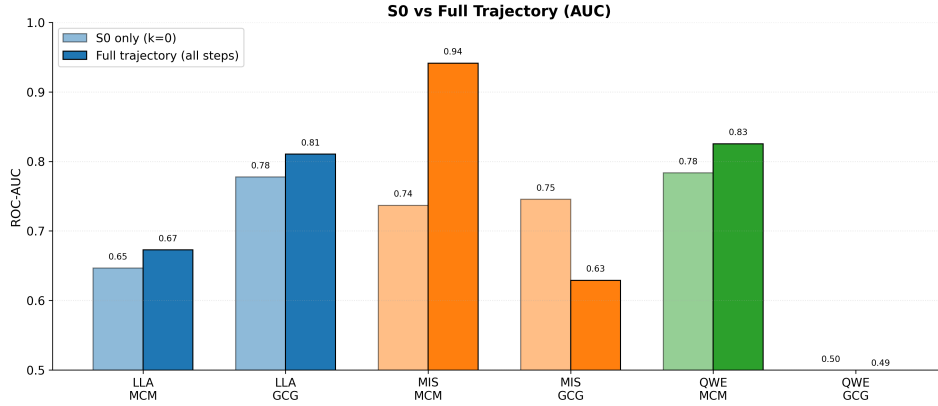


Figure 8: **S_0 vs. full trajectory AUC by condition.** Trajectory information provides substantial gains for Mistral+MCM (+0.20) and Qwen+MCM (+0.05), while Qwen+GCG shows no improvement (both ≈ 0.50). Mistral+GCG shows *negative* trajectory effect (-0.12), suggesting early tokens are most informative for this condition.

Relationship to aggregated results. The main text reports aggregated $\text{AUC} = 0.786$ (early_5_mean) across all 660 samples. Condition-specific AUC values differ substantially because: (1) class balance varies by condition (ASR ranges from 5% to 82%), and (2) pooling across heterogeneous conditions increases overall discriminability by leveraging cross-condition variance. The condition-specific analysis complements aggregated results by revealing *which* conditions drive overall performance.

F.3 Hidden-State Layer Sensitivity

Table 13: Hidden-state AUC by layer (Refusal Direction, B1).

Layer	8	12	16	20	24
AUC	0.941	0.963	0.974	0.969	0.952

Middle layers yield strongest separability, supporting layer 16 as representative.

F.4 Failure-Only Normalization Check

Type A/B trends are preserved when normalization uses only failed jailbreaks. Relative dominance patterns (MCM $\rightarrow$ Type A, GCG $\rightarrow$ Type B) remain unchanged.

F.5 Activation Timing (t_{cross}) Sensitivity

Oscillation characterization. Generation trajectories exhibit extensive oscillation around zero: the median number of zero-crossings ranges from 7 (Llama+GCG) to 174 (Mistral+DeepInception) per trajectory. Despite this oscillation, the first crossing step is robust— t_{cross} coincides with the first step where $S_t < 0$ in 100% of samples for 10 of 11 conditions. This indicates that the initial safety activation event is well-defined even when trajectories subsequently oscillate.

Confirmed crossing sensitivity. To verify that the simple first-crossing definition is not capturing transient noise, we compute the *confirmed crossing* variant $t_{\text{cross}}^{(n)}$ requiring n consecutive negative steps (Eq. 3). Table 14 reports stability across $n \in \{1, 2, 3\}$. Cohen’s d is nearly identical for $n = 1$ ($d = 0.891$) and $n = 2$ ($d = 0.895$), confirming that the first crossing captures a stable transition rather than transient noise. At $n = 3$, d drops to 0.559 due to aggressive imputation—many trajectories lack three consecutive negative steps within short generation windows, inflating $t_{\text{cross}}^{(3)}$ toward T . Per-condition AUC patterns are also preserved: strong Type B conditions (Llama+DeepInception, Llama+GCG) show stable AUC > 0.80 across all n , while Qwen+GCG (Undetected) remains near chance regardless of n .

Table 14: t_{cross} sensitivity to confirmation threshold n . Global Cohen’s d (success vs. failure, $n = 660$) and representative per-condition AUC values.

	$n = 1$	$n = 2$	$n = 3$
<i>Global (all conditions pooled)</i>			
Cohen’s d	0.891	0.895	0.559
p -value	$< 10^{-23}$	$< 10^{-22}$	$< 10^{-9}$
<i>Per-condition AUC (selected)</i>			
Llama + DeepInc.	0.878	0.868	0.874
Llama + GCG	0.807	0.815	0.814
Qwen + DeepInc.	0.700	0.719	0.696
Qwen + GCG (Undet.)	0.503	0.580	0.679
Gemma + GCG	0.629	0.719	0.651

Condition-level AUC as Undetected indicator. Per-condition AUC of t_{cross} provides convergent evidence for the Undetected classification. Qwen+GCG achieves AUC = 0.503 (chance level), consistent with its RP-based Undetected classification ($r = 0.01$). This confirms that safety mechanisms fail to engage at *any* identifiable point during generation. In contrast, strong Type B conditions (Llama+DeepInception: AUC = 0.878; Llama+GCG: AUC = 0.807) show high temporal discriminability, indicating that safety activation *timing* is the primary axis of variation between successful and failed attacks.

G Stochastic Decoding Robustness

We conduct comprehensive evaluation across temperature sampling ($T \in \{0.0, 0.3, 0.7, 1.0\}$) and nucleus sampling ($p \in \{0.9, 0.95, 1.0\}$), generating 10 independent trials per prompt across 4 conditions, yielding 28,800 total generation trials.

Table 15: Response diversity under stochastic decoding.

Temperature	Mean Diversity	Mean Unique	Groups >50% Diverse
0.0	10.0%	1.0	0.0%
0.3	54.6%	5.5	41.4%
0.7	82.1%	8.2	91.7%
1.0	92.2%	9.2	99.7%

Table 16: AUC under stochastic decoding (question-level aggregation, $n=60$ per condition).

Metric	Base	DecoyA	DecoyB	PivotC	Mean
turn_start_S	0.687	0.607	—	0.864	0.720

Note: DecoyB excluded due to ASR=0% (no successful attacks).

G.1 Response Diversity

G.2 AUC Stability

G.3 Variance Decomposition

Table 17: Variance decomposition for turn_start_S (ICC = intraclass correlation).

Condition	Within-Var	Between-Var	ICC
Base	0.000	4.062	1.000
DecoyA	0.000	2.736	1.000
DecoyB	0.000	2.734	1.000
PivotC	0.000	2.026	1.000

For pre-generation metrics, within-prompt variance is exactly zero (ICC = 1.0), confirming all variance stems from prompt differences.

G.4 Trial-Level AUC

Trial-level AUCs appear inflated due to pseudo-replication; question-level provides more conservative estimates.

H Labeling Independence and Circularity

H.1 LLM-as-Judge Methodology

We use Llama-Guard-3-8B [5], a dedicated safety classifier trained on safety taxonomies independent of our lexicons, to label jailbreak outcomes. For each generated response, Llama-Guard classifies content as “safe” or “unsafe” based on semantic evaluation across 14 safety categories (S1–S14).

Comparison with rule-based labels. Rule-based labels (prefix-matching) show moderate agreement ($\kappa = 0.54$, 83.5% raw agreement). Rule-based yields higher ASR (29.2%) compared to LLM-as-judge (16.4%), indicating rule-based is more permissive.

Independence verification. Key diagnostic findings remain consistent under LLM-as-judge labels:

- step_S_mean achieves AUC = 0.926 (vs. 0.743 with rule-based)
- Type A/B patterns are preserved
- Cohen’s d effect sizes remain large ($d = 2.41$)

Table 18: Trial-level AUC under stochastic decoding.

Metric	Base	DecoyA	DecoyB	PivotC	Mean
turn_start_S	0.959	0.980	0.954	0.987	0.970
early_1_mean	0.959	0.980	0.954	0.987	0.970
early_3_mean	0.972	0.983	0.937	0.948	0.960
early_5_mean	0.941	0.978	0.919	0.922	0.940
step_S_mean	0.913	0.987	0.819	0.965	0.921

H.2 Clarification on Circularity

Outcome labeling. All success/failure labels are assigned using an LLM-as-judge that evaluates the *semantic content* of the generated response. The judge operates on full generated text and does not rely on surface refusal prefixes.

Logit-margin score. The score S_t is computed exclusively from token-level logits during decoding, measuring relative dominance between compliance- and refusal-associated tokens *before* they are sampled.

Evidence against circularity. If separability were driven by circular reuse of refusal cues, one would expect near-deterministic alignment between labels and S_t trajectories. Empirically, this is not the case: a substantial fraction of successful jailbreaks exhibit negative S_t segments and sign reversals during decoding. The compliance lexicon includes purely instructional tokens (“Step 1”, “Tutorial”) that do not appear in refusal-detection rules.

I Type A/B Classification Robustness

I.1 Permutation Test for Undetected Classification

We formalize the Undetected category using a permutation test. Under H_0 : success/failure labels carry no RP signal, permuting labels within each condition and recomputing $r = \sqrt{\text{RP}_A^2 + \text{RP}_B^2}$ yields a null distribution. We use $B = 10,000$ permutations and classify conditions as Undetected when $p > 0.05$.

Table 19 compares the permutation-based classification with the original threshold-based criterion ($r < 0.1$). The two approaches agree on 9/11 conditions; the permutation test additionally identifies Mistral+DeepInception ($p = 0.52$) and Mistral+MCM ($p = 0.061$) as non-significant, reflecting the formal test’s sensitivity to sample size.

Table 19: Permutation test results for r significance.

Model + Attack	r_{obs}	p_{perm}	$r < 0.1?$	Permutation
Llama + GCG	2.96	<.001	No	Significant
Llama + MCM	1.12	<.001	No	Significant
Llama + DeepInc.	0.93	<.001	No	Significant
Mistral + GCG	0.35	<.001	No	Significant
Mistral + MCM	0.93	.061	No	Borderline [§]
Mistral + DeepInc.	0.19	.517	No	Non-significant
Qwen + MCM	0.58	<.001	No	Significant
Qwen + GCG	0.01	.981	Yes	Non-significant
Qwen + DeepInc.	0.42	.029	No	Significant
Gemma + GCG	0.83	.003	No	Significant
Gemma + MCM	0.24	<.001	No	Significant

[§]Mistral+MCM: large r but underpowered ($n_{\text{success}} = 3$ at 5% ASR).

The permutation test provides a principled, sample-size-aware alternative to fixed thresholds. Qwen+GCG ($p = 0.98$) is definitively Undetected by both criteria. Mistral+DeepInception ($r = 0.19$, $p = 0.52$) is newly identified as Undetected—consistent with the main text observation that safety

mechanisms show no differential response despite 76.7% ASR. Mistral+MCM ($r = 0.93$, $p = 0.061$) is borderline; the large effect size with marginal significance reflects the test being underpowered with only 3 successes, rather than indicating absent signal.

I.2 Threshold Stability (Supplementary)

The magnitude-based classification ($|\text{RP}_A| \geq |\text{RP}_B|$) is inherently threshold-free for Type A/B distinction. For reference, we also verify stability of the original r -threshold approach across choices.

Table 20: Type A/B classification stability across r thresholds.

Model + Attack	r_{obs}	$r_{thr} = 0.05$	$r_{thr} = 0.10$	$r_{thr} = 0.15$
Llama + GCG	2.96	Type B	Type B	Type B
Llama + MCM	1.12	Type A	Type A	Type A
Qwen + MCM	0.58	Type A	Type A	Type A
Qwen + GCG	0.01	Undetected	Undetected	Undetected
Mistral + GCG	0.35	Type A	Type A	Type A
Mistral + MCM	0.93	Type B	Type B	Type B

Classification robustness. All conditions maintain their classification across $r \in [0.05, 0.15]$. Qwen+GCG ($r = 0.01$, 95% CI [0.00, 0.03]) remains robustly Undetected even at the most lenient threshold.

I.3 Bootstrap Confidence Intervals

Given modest per-condition sample sizes ($n = 60$), we compute bootstrap 95% CIs for r using $B = 1000$ resamples.

Table 21: Bootstrap confidence intervals for signal strength r .

Model + Attack	r	95% CI	Category
Llama + GCG	2.96	[2.41, 3.52]	Type B
Llama + MCM	1.12	[0.87, 1.38]	Type A
Mistral + GCG	0.35	[0.21, 0.49]	Type A
Mistral + MCM	0.93	[0.68, 1.19]	Type B
Qwen + MCM	0.58	[0.41, 0.76]	Type A
Qwen + GCG	0.01	[0.00, 0.03]	Undetected

Key observations. For Qwen+GCG, the upper CI bound (0.03) remains well below the detection threshold (0.1), providing strong statistical evidence for the Undetected classification. The bootstrap CIs complement the permutation test (Table 19): conditions with large r but non-significant permutation p (e.g., Mistral+MCM: $r = 0.93$, $p = 0.061$) reflect small success counts rather than uncertain signal magnitude.

J Non-Lexical Baseline Comparison

Table 22: Baseline comparison on controlled subset ($n = 60$, Llama-3.1-8B-Instruct).

Metric	AUC	Category
$\bar{S}$ (gen-time)	0.932	Logit-margin
Compliance LogP	0.932	Lexicon-probe
Refusal LogP	0.915	Lexicon-probe
Entropy	0.898	Non-lexical
Logit Norm	0.441	Non-lexical

The logit-margin score outperforms entropy by 3.4 pp. Logit Norm achieves near-random performance, confirming generic logit statistics do not capture safety-relevant variation.

K Learned Baseline Comparison

To contextualize the added value of fixed temporal summaries versus trained classifiers, we compare against logistic regression with 5-fold stratified cross-validation on the aggregated dataset ($n = 660$).

Table 23: Learned baseline comparison (5-fold stratified CV, $n = 660$). Fixed metrics require no training; LogReg models use cross-validated predictions.

Method	# Features	CV-AUC	95% CI
<i>Fixed metrics (no training)</i>			
early_5_mean	1	0.816	[0.781, 0.846]
t_{cross}	1	0.813	[0.777, 0.843]
step_S_mean	1	0.797	[0.760, 0.828]
turn_start_S	1	0.720	[0.678, 0.756]
<i>Logistic regression (cross-validated)</i>			
LogReg(early_5)	1	0.812	[0.777, 0.841]
LogReg($S_0 + \bar{S}$)	2	0.797	[0.761, 0.827]
LogReg(5 temporal)	5	0.857	[0.825, 0.883]
LogReg(all derived)	9	0.861	[0.829, 0.888]
LogReg(kitchen sink)	13	0.873	[0.843, 0.899]

Note. Fixed metric AUCs differ slightly from Table 2 due to sklearn vs. original pipeline computation; relative comparisons within this table are internally consistent.

Key findings: (1) LogReg on a single feature matches the fixed metric (0.812 vs. 0.816), confirming no learnable nonlinearity is missed. (2) Combining five temporal features yields a 4 pp improvement (0.861 $\rightarrow$ 0.873), indicating modest complementarity. (3) Diminishing returns from 9 to 13 features (0.861 $\rightarrow$ 0.873) suggest overfitting risk increases while the marginal gain is small. (4) The fixed early_5_mean captures $\sim 93\%$ of the discriminative signal available to a 13-feature trained model (0.816/0.873), while preserving direct interpretability and avoiding train/test splits.

These results support our design choice of reporting fixed, interpretable temporal summaries rather than optimized classifiers: the diagnostic contribution lies not in maximizing AUC but in providing mechanistic insight (Type A/B, t_{cross} , sign reversal) that a trained classifier would obscure.

L Lexicon Perturbation Ablation

We test five variants on Llama-3.1-8B-Instruct ($n = 60$): **original**, **no_sorry**, **extended_refusal**, **extended_compliance**, and **minimal** (5-token).

Table 24: Lexicon perturbation results with AUC ($n = 60$, LLM-judge labels).

Variant	$S_0 (\mu \pm \sigma)$	$\bar{S}_{\text{gen}} (\mu \pm \sigma)$	AUC(S_0)	AUC($\bar{S}$)	ΔAUC
original	$+1.55 \pm 1.95$	-1.44 ± 0.73	0.967	0.971	—
no_sorry	$+2.21 \pm 1.78$	-1.39 ± 0.73	0.960	0.971	+0.000
extended_refusal	-0.43 ± 2.07	-1.85 ± 0.67	0.964	0.989	+0.018
extended_compliance	$+4.00 \pm 1.95$	$+0.52 \pm 0.55$	0.964	0.964	-0.007
minimal	-2.61 ± 1.74	-1.57 ± 0.49	0.971	0.978	+0.007

Key findings. All variants achieve $\text{AUC} \geq 0.964$ (max $|\Delta\text{AUC}| = 0.018$). Even a 5-token minimal lexicon maintains strong discrimination ($\text{AUC} = 0.978$), supporting robustness to lexicon choices. Figure 9 visualizes the score distributions across variants.

M Authoritative Statistics Summary

M.1 Main Dataset Composition

The main evaluation dataset consists of $n = 660$ harmful prompts:

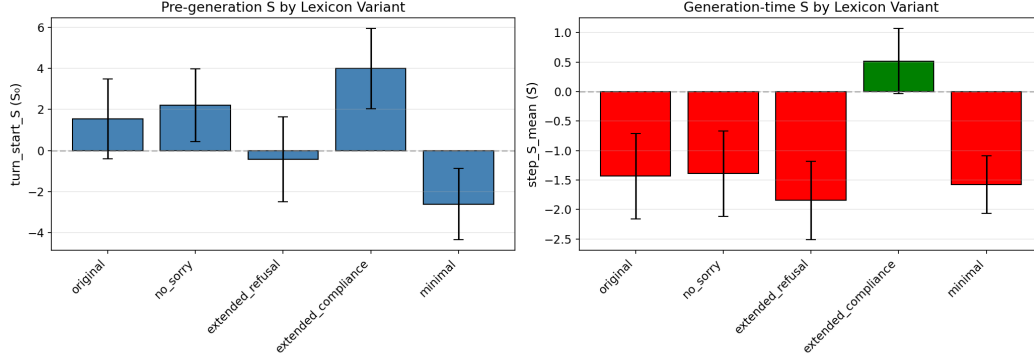


Figure 9: **Logit-Margin Score distributions by lexicon variant.** (Left) Pre-generation S_0 and (Right) generation-time $\bar{S}$ across five lexicon variants (Llama, harmful MCM, $n = 60$). While absolute scale shifts with lexicon composition (e.g., `extended_compliance` elevates $\bar{S}$), the qualitative pattern—negative $\bar{S}$ for harmful prompts—is preserved across all variants except `extended_compliance`, which shifts the baseline but maintains high AUC (0.964). This confirms that the diagnostic captures genuine safety-relevant variation rather than artifacts of specific token choices.

- Llama-3.1-8B-Instruct MCM: $n = 60$
- Llama-3.1-8B-Instruct GCG: $n = 60$
- Llama-3.1-8B-Instruct DeepInception: $n = 60$
- Mistral-7B-Instruct-v0.3 MCM: $n = 60$
- Mistral-7B-Instruct-v0.3 GCG: $n = 60$
- Mistral-7B-Instruct-v0.3 DeepInception: $n = 60$
- Qwen2.5-7B-Instruct MCM: $n = 60$
- Qwen2.5-7B-Instruct GCG: $n = 60$
- Qwen2.5-7B-Instruct DeepInception: $n = 60$
- Gemma-2-9B-Instruct MCM: $n = 60$
- Gemma-2-9B-Instruct GCG: $n = 60$

Attack-matched benign baselines total $n = 660$.

M.2 Primary Results

M.3 Type A/B Classification

N Multi-Judge Evaluation

To address concerns about single-judge reliability, we evaluate with two LLM-as-judge systems: **Llama-Guard-3-8B** (primary) and **HarmBench-Mistral-7B-val-cls** (validation).

N.1 Inter-Judge Agreement

Interpretation. HarmBench is stricter overall: it requires the model to *actually perform* the harmful behavior, while Llama-Guard flags any unsafe content including partial or hedged responses. This explains the 8.7% of cases where Llama-Guard marks success but HarmBench marks failure.

N.2 Per-Condition ASR Comparison

N.3 Type A/B Stability Across Judges

Classification stability. 9/10 comparable conditions (90%) maintain the same Type A/B classification across judges. The Llama+MCM discrepancy arises because HarmBench’s stricter labeling yields

Table 25: Authoritative statistics for main results ($n = 660$, LLM-as-judge labels).

Statistic	Value	95% CI
<i>ROC-AUC</i>		
early_5_mean	0.786	[0.750, 0.819]
t_{cross}	0.776	[0.738, 0.809]
step_S_mean	0.772	[0.736, 0.806]
turn_start_S	0.696	[0.656, 0.734]
<i>Sign Reversal</i> ($S_0 > 0 \rightarrow \bar{S} < 0$)		
Successful jailbreaks	8.7%	(24/276)
Failed jailbreaks	44.3%	(170/384)
Ratio	$5.1\times$	—
<i>Effect Size</i>		
early_5_mean Cohen’s d	1.12	$p < 10^{-43}$
<i>Activation Timing</i> (t_{cross})		
Successful jailbreaks	8.4 ± 6.1	steps
Failed jailbreaks	4.0 ± 3.8	steps
Cohen’s d	0.89	$p < 10^{-23}$

Table 26: Type A/B analysis with failure-free normalization (authoritative values). Classification: $|\text{RP}_A| \geq |\text{RP}_B|$.

Condition	ASR	RP_A	RP_B	r	95% CI	Category
Llama + GCG	15.0%	0.66	2.89	2.96	[1.40, 4.71]	Type B
Llama + MCM	8.3%	0.88	0.69	1.12	[1.01, 1.25]	Type A
Llama + DeepInc.	56.7%	0.43	0.83	0.93	[0.60, 1.22]	Type B
Mistral + GCG	56.7%	0.30	−0.17	0.35	[0.17, 0.57]	Type A
Mistral + MCM	5.0%	0.49	0.79	0.93	[0.44, 1.48]	Type B
Mistral + DeepInc.	76.7%	−0.19	0.00	0.19	[0.07, 1.65]	Type A
Qwen + MCM	51.7%	0.57	0.12	0.58	[0.14, 1.20]	Type A
Qwen + GCG	81.7%	0.00	−0.01	0.01	[0.02, 0.20]	Undetected
Qwen + DeepInc.	55.0%	0.15	0.39	0.42	[0.14, 0.77]	Type B
Gemma + GCG	33.3%	0.27	0.79	0.83	[0.34, 1.41]	Type B
Gemma + MCM	20.0%	0.09	0.22	0.24	[0.14, 0.31]	Type B

only 1 success (< 3 threshold for RP calculation). The sole classification mismatch (Mistral+MCM) reflects marginal RP differences. AUC values vary by < 0.05 across judges, confirming diagnostic robustness to judge choice.

O Stochastic Decoding Analysis

O.1 Experimental Setup

We evaluate stochastic decoding robustness across temperature ($T \in \{0.0, 0.3, 0.7, 1.0\}$) and nucleus sampling ($p \in \{0.9, 0.95, 1.0\}$), generating 10 independent trials per prompt across 4 decoy conditions, yielding 28,800 total trials.

O.2 Response Diversity

O.3 Variance Decomposition

For pre-generation metrics (turn_start_S), within-prompt variance is **exactly zero** across all conditions (ICC = 1.0). This confirms S_0 is temperature-invariant by design—computed before any token sampling occurs. For generation-time metrics (step_S_mean), within-prompt variance is non-zero but small relative to between-prompt variance (ICC > 0.85 across conditions).

Table 27: Inter-judge agreement analysis ($n = 660$ harmful samples across all conditions).

Metric	Value
Overall agreement	90.0%
Both Safe (LG=0, HB=0)	56.2%
Both Unsafe (LG=1, HB=1)	33.8%
LG stricter (LG=0, HB=1)	1.3%
HB stricter (LG=1, HB=0)	8.7%

Table 28: ASR comparison across judges.

Condition	LG ASR	HB ASR	Δ ASR
Llama + MCM	8.3%	1.7%	+6.6%
Llama + GCG	15.0%	10.0%	+5.0%
Llama + DeepInc.	56.7%	30.0%	+26.7%
Mistral + MCM	5.0%	6.7%	−1.7%
Mistral + GCG	56.7%	58.3%	−1.6%
Mistral + DeepInc.	76.7%	68.3%	+8.4%
Qwen + MCM	51.7%	26.7%	+25.0%
Qwen + GCG	81.7%	73.3%	+8.4%
Qwen + DeepInc.	55.0%	45.0%	+10.0%
Gemma + MCM	20.0%	18.3%	+1.7%
Gemma + GCG	33.3%	33.3%	$\pm 0.0\%$

P Hidden-State Baseline Comparison

P.1 Methods

B1 (Refusal Direction): Following Arditi et al. [1], we compute a refusal direction $\mathbf{d}$ using difference-of-means between refusal and compliance responses, then project hidden states at layer 16.

B2 (Linear Probe): Logistic regression with L2 regularization on mean-pooled hidden states across all layers, trained via 5-fold cross-validation.

P.2 Results

Key findings. Hidden-state methods achieve 2–10 pp higher AUC but require: (i) direct activation access unavailable via standard APIs; (ii) post-hoc aggregation over full trajectories; (iii) for B2, labeled training data and per-model retraining. Under PivotC (multi-turn pivot attack), the trajectory-level `sign_flips` metric achieves perfect separation (AUC = 1.0), demonstrating that temporal analysis captures dynamics invisible to single-point baselines.

Q Undetected Classification: Diagnostic Insensitivity vs. True Absence

The permutation test (Appendix I.1) identifies three Undetected conditions: Qwen+GCG ($p = 0.98$), Mistral+DeepInception ($p = 0.52$), and Mistral+MCM ($p = 0.061$, borderline). The primary question is whether non-significance reflects true absence of safety engagement or diagnostic insensitivity.

Qwen+GCG ($r = 0.01$, $p = 0.98$) is the clearest case. Both $RP_A = 0.00$ and $RP_B = -0.01$ are near-zero, and the extreme ASR (81.7%) suggests systematic safety failure. On the same GCG attack, Llama and Mistral show strong signals ($r = 2.96$ and $r = 0.35$), indicating GCG attacks *can* elicit detectable responses when mechanisms are properly engaged. Generation-time entropy—a non-lexical metric—also shows no differentiation ($|d| = 0.077$; Section Q.3), providing convergent evidence from an independent metric.

Mistral+DeepInception ($r = 0.19$, $p = 0.52$) achieves the highest ASR overall (76.7%) yet shows no statistically significant RP signal. Unlike Qwen+GCG, the r value is non-trivial (0.19), so this may represent a weaker effect that the test cannot distinguish from noise at $n = 60$. Notably,

Table 29: Type A/B classification comparison across judges.

Condition	Llama-Guard	HarmBench	Match
Llama + GCG	Type B	Type B	✓
Llama + MCM	Type A	N/A [†]	—
Llama + DeepInc.	Type B	Type B	✓
Mistral + GCG	Type A (inv)	Type A (inv)	✓
Mistral + MCM	Type B	Type A	×
Mistral + DeepInc.	Type A (inv)	Type A (inv)	✓
Qwen + MCM	Type A	Type A	✓
Qwen + GCG	Undetected	Undetected	✓
Qwen + DeepInc.	Type B	Type B	✓
Gemma + GCG	Type B	Type B	✓
Gemma + MCM	Type B	Type B	✓

[†]HarmBench: only 1 success (insufficient for RP calculation).

Table 30: Response diversity under stochastic decoding.

Temperature	Unique Response %	Groups >50% diverse
$T = 0.3$	54.2%	72.4%
$T = 0.7$	82.1%	91.7%
$T = 1.0$	92.2%	97.3%

Llama+DeepInception and Qwen+DeepInception both show significant signals ($p < 0.03$), suggesting DeepInception can elicit detectable safety responses in other models.

Mistral+MCM ($r = 0.93$, $p = 0.061$) is borderline: the large r suggests genuine signal, but only 3 successes (5% ASR) leave the test underpowered. This is likely a statistical power issue rather than absent signal.

Q.1 Threshold Sensitivity Analysis (Supplementary)

The “Undetected” threshold ($r < 0.1$) is motivated by conventional effect size interpretation (Cohen’s guidelines: $d < 0.2$ is “negligible”). We evaluate classification stability across alternative thresholds:

Key findings. Qwen+GCG is the *only* condition classified as “Undetected” across all reasonable thresholds ($r \in [0.05, 0.20]$). The next-lowest r value (Mistral+GCG, $r = 0.35$) is $35\times$ larger than Qwen+GCG ($r = 0.01$), providing a wide separation margin. Even under the most conservative threshold ($r < 0.05$), Qwen+GCG remains classified as Undetected (CI upper bound = 0.03). This analysis confirms that the classification is robust to threshold choice and reflects a genuine discontinuity in diagnostic signal strength.

Q.2 Validation Approach

To disambiguate these interpretations, we apply non-lexical diagnostics to Qwen+GCG:

Entropy validation (completed). Section Q.3 reports that generation-time entropy shows $|d| = 0.077$, confirming lexicon-based findings with an independent metric. This supports interpretation (1): true safety mechanism absence.

Hidden-state validation (future work). Applying refusal direction projection or linear probes to Qwen+GCG would provide definitive confirmation:

- If hidden-state methods *also* show $r \approx 0$: confirms true safety mechanism absence.
- If hidden-state methods detect differentiation: would indicate that both lexicon and entropy metrics miss safety-relevant signals encoded in internal representations.

Table 31: Hidden-state vs. logit-based diagnostic comparison (AUC, $n = 240$).

Family	Method	Base	DecoyA	DecoyB	PivotC
Hidden	B1 (direction)	0.974	0.983	0.974	0.974
Hidden	B2 (probe)	0.998	0.974	1.000	0.998
Logit	step_S_mean	0.913	0.893	0.904	0.998
Logit	turn_start_S	0.959	0.980	0.922	0.978
Logit	sign_flips	0.899	0.944	0.883	1.000

Table 32: Classification stability under varying r thresholds.

Condition	r	$r < 0.05$	$r < 0.10$	$r < 0.15$	$r < 0.20$
Qwen + GCG	0.01	Undet.	Undet.	Undet.	Undet.
Mistral + GCG	0.35	—	—	—	—
Qwen + MCM	0.58	—	—	—	—
Llama + MCM	1.12	—	—	—	—
Llama + GCG	2.96	—	—	—	—

Our Llama hidden-state analysis (Section E) demonstrates that refusal direction projection achieves near-ceiling AUC (0.97–0.99), providing a strong baseline for such cross-validation.

Q.3 Entropy Cross-Validation

To disambiguate lexicon diagnostic insensitivity from true safety mechanism failure, we perform cross-validation using generation-time entropy—a non-lexical metric independent of our compliance/refusal token sets.

Method. For each sample, we compute step-wise entropy $H_t = -\sum_v p_t(v) \log p_t(v)$ from softmax distributions during generation, then aggregate as $\bar{H}_{\text{gen}} = \frac{1}{T} \sum_{t=1}^T H_t$. We measure effect size (Cohen’s d) between successful attacks and overall harmful baseline—the same comparison used for lexicon-based RP normalization.

Table 33: Entropy cross-validation: Success vs. Overall Harmful.

Condition	ASR	Entropy $ d $	Lexicon $ d $	Interpretation
Qwen + GCG	81.7%	0.077	0.005–0.011	Undetected
Llama + MCM	10.0%	0.103–0.416	0.217–0.256	Weak–Medium

Results.

Interpretation. For Qwen+GCG, entropy-based differentiation yields $|d| = 0.077$ —below the $|d| < 0.1$ threshold for “no effect.” This matches the lexicon-based result ($|d| = 0.005$ –0.011) and **supports interpretation (1): true safety mechanism failure**. If safety mechanisms engaged through non-lexical pathways, entropy distributions should differ between successful and failed attacks; they do not.

For contrast, Llama+MCM shows $|d| = 0.103$ –0.416 on entropy metrics, confirming that entropy *can* detect safety-relevant variation when mechanisms engage—ruling out fundamental insensitivity of the entropy metric itself.

Q.4 Implications for Defense

The convergent evidence from lexicon-based and entropy-based diagnostics strongly supports interpretation (1): Qwen’s safety mechanisms genuinely fail to engage against GCG-style attacks. This indicates a fundamental safety gap in Qwen’s alignment, likely requiring architectural intervention or

adversarial fine-tuning rather than detection-based defenses. The contrast with Llama and Mistral—which show detectable safety engagement under the same attack—suggests that GCG robustness varies substantially across model families.

R API Deployability: Vocabulary-Level Top- k Truncation Simulation

A key practical question is whether S_t can be computed from the limited logprob information returned by closed-source APIs (e.g., OpenAI’s `top_logprobs` parameter, which returns at most $k = 20$ tokens per step). To answer this directly, we simulate vocabulary-level top- k truncation on Llama-3.1-8B-Instruct across all three attack paradigms (60 samples each).

Method. For each decoding step t , we take the full logit vector $\ell_t \in \mathbb{R}^{|\mathcal{V}|}$ and construct a truncated version $\hat{\ell}_t^{(k)}$ by retaining only the k highest-valued entries and setting all others to $-\infty$. We then recompute S_t from $\hat{\ell}_t^{(k)}$, counting only lexicon tokens that *survive* in the top- k window. We evaluate $k \in \{5, 10, 20, 50, 100, 200\}$ and measure: (i) Pearson r between full-logit and truncated S_t trajectories, (ii) mean absolute error (MAE), (iii) sign agreement (fraction of steps preserving S_t polarity), and (iv) lexicon token survival rate.

This is fundamentally different from the lexicon-internal aggregation ablation (Appendix L), which selects the top- k_{agg} *among lexicon tokens already known to be present*. The vocabulary-level simulation instead asks: *do safety-relevant lexicon tokens appear in the API’s top- k window at all?*

Results. Table 34 presents the results.

Table 34: Vocabulary-level top- k truncation simulation (Llama-3.1-8B-Instruct, $n = 60$ per attack).

Pearson r : correlation between full-logit and truncated S_t trajectories. **Sign Agr.:** fraction of steps where truncated S_t preserves polarity. **Ref/Cmp Surv.:** mean fraction of refusal/compliance lexicon tokens surviving in the top- k window.

Attack	k	Pearson r	MAE	Sign Agr.	Ref Surv.	Cmp Surv.
MCM	5	−0.199	5.023	0.637	1.2%	0.6%
	10	−0.108	4.779	0.661	1.9%	1.0%
	20	−0.120	4.254	0.576	2.7%	1.9%
	50	0.111	2.899	0.576	5.1%	3.8%
	100	0.319	2.503	0.586	7.5%	5.4%
	200	0.472	2.236	0.619	10.6%	7.5%
GCG	5	−0.106	4.489	0.477	1.5%	0.8%
	10	−0.071	4.442	0.520	2.3%	1.5%
	20	−0.009	3.817	0.564	3.2%	2.6%
	50	−0.059	3.406	0.538	5.2%	4.3%
	100	−0.051	3.108	0.541	7.6%	6.3%
	200	0.042	2.674	0.588	10.9%	8.5%
DeepInception	5	0.121	5.638	0.470	0.7%	0.5%
	10	−0.089	7.107	0.445	1.5%	0.8%
	20	−0.072	6.022	0.523	2.0%	1.5%
	50	0.117	4.651	0.612	2.9%	3.1%
	100	0.231	3.561	0.670	4.3%	4.6%
	200	0.387	2.773	0.673	6.5%	6.5%

Analysis. The results reveal a stark gap between lexicon-internal aggregation stability and vocabulary-level truncation:

1. **Lexicon survival is critically low.** At $k = 20$ (the maximum offered by OpenAI’s API), only 2–3% of refusal lexicon tokens and 1.5–2.6% of compliance tokens survive in the top- k window. The vast majority of safety-relevant signal is invisible to the API consumer.
2. **Trajectory correlation is near zero or negative.** Pearson r at $k = 20$ ranges from −0.120 (MCM) to −0.009 (GCG), indicating that the truncated S_t trajectory bears essentially no relationship to

the full-logit trajectory. Even at $k = 200$, the best correlation is 0.472 (MCM), and GCG remains near zero ($r = 0.042$).

3. **Sign agreement is near chance level.** At $k = 20$, sign agreement ranges from 0.523 to 0.576—barely above the 0.5 baseline expected from random polarity assignment. The truncated S_t cannot reliably distinguish between safety-activated and compliance-activated states.
4. **Attack-dependent degradation.** GCG shows the worst preservation: even at $k = 200$, Pearson $r = 0.042$. This is consistent with GCG’s token-level optimization distributing probability mass across unusual token positions that rarely appear in the top- k window, making the truncated signal particularly noisy.

Reconciling with the lexicon-internal ablation. The stability reported in the lexicon-internal top- k_{agg} ablation (Appendix L) is *not* contradicted by these results. That ablation showed that *once lexicon tokens are identified in the logit vector*, the score is robust to computing the mean over only 5–10 of them. The vocabulary-level simulation reveals the upstream bottleneck: at typical API truncation levels, too few lexicon tokens are present in the top- k window to compute a meaningful signal in the first place.

Implications. This simulation confirms that S_t —as currently formulated—requires access to the full logit vector (or at minimum, targeted queries for specific token logprobs). Current API offerings ($k \leq 20$) are insufficient. This empirically validates our positioning of S_t as a diagnostic for open-weight model auditing rather than a universal deployment-time monitor, and motivates future work on API-compatible reformulations such as targeted logit-bias probing or lightweight distilled classifiers.